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基於移動窗口和反向殘差互換器之

3-D 漩渦 Transformer 於自動乳房超音波腫瘤診斷

A 3-D SWIRL Transformer based on Swin and

Inverted Residual Switch for Tumor Diagnosis in

Automated Breast Ultrasound Image

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本論文係簡雅慧君（學號 R06921016）在國立臺灣大學電機資訊學院生醫電子與資訊學研究所完成之碩士學位論文，於民國 111 年 8 月 9 日承下列考試委員審查通過及口試及格，特此證明

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致謝

經歷了些許波折、體驗疫情影響後的研究生生活後，碩士論文也即將進入尾聲。在這段漫長的時間了，多虧了很多人的幫助才能重新調整好自己的狀態、繼續向前邁進。首先，感謝老師願意收我為研究生，並且提供充分的軟硬體設備讓我可以順利進行實驗，也讓我的研究題目剛好回到小時候的初衷，儘管離去的家人已經來不及，但還是希望能有機會幫助到未來乳癌方面的研究進展。另外，還要感謝實驗室的學長姐、同學以及學弟妹們，讓我能很快融入大家，一起討論研究方法、準備課業、吃飯、聊天、慶生，遠端進行研究的時候也能協助處理斷線或硬體問題。

最後，也要感謝我的家人以及同等重要的人，在最難過的時候一起承受和分擔失去的傷痛，悲傷漸漸地隨著時間沖淡，但還是會遺憾離開的人已走向終點，沒有機會當面分享這段時間累積的成果和喜悅。即將進入下一個階段，期待著未來的可能性，但也對之後的不確定性感到忐忑不安，期許我能在社會既定印象中活出自我，對於自我原則有一定堅持，也能包容他人各種觀念。

摘要

乳癌是主要造成死亡的癌症之一，及早診斷與治療可以大幅提升乳癌病患的生存率。自動乳房超音波(automated breast ultrasound, ABUS)是常見的乳房檢測技術，提供乳房組織的三維資訊。然而大量二維切片構成的乳房超音波影像讓醫生花費冗長的時間審閱，並且延誤診斷惡性腫瘤。近年來開發了基於深度學習的電腦輔助診斷(computer-aided diagnosis, CADx)系統來自動提取特徵並加速診斷流程。因此本研究提出一個基於 Transformer 和卷積神經網路(Convolutional Neural Network, CNN)的電腦輔助診斷系統以診斷乳房超音波影像。提出的系統利用自注意力機制來獲得全局影像關係，且加入反向殘差卷積架構來提取局部影像資訊。

本系統包含影像前處理、腫瘤切割和腫瘤分類。在影像前處理，會提取腫瘤區域並調整至固定影像大小，然後利用直方圖均衡化增強影像對比度。接著，調整大小後的影像會透過腫瘤分割模型產生腫瘤遮罩。最後，調整大小後的腫瘤影像、增強後的腫瘤影像和腫瘤遮罩會作為所提出的三維漩渦 Transformer 腫瘤分類模型的輸入並判斷腫瘤良惡性。而提出的模型是由移動窗口自注意力機制和反向殘差互換器所建構來結合全局和局部特徵。根據實驗結果，本研究提出的電腦輔助診斷系統可達到 89.9% 的正確率、89.8% 的靈敏性、89.9% 的特異性和 0.9401 的曲線下面積。實驗結果顯示提出的系統可以有效輔助醫生更準確的診斷乳癌。

關鍵詞：乳癌、自動乳房超音波、電腦輔助診斷系統、卷積神經網路、自注意力機制

Abstract

Breast cancer is one of the leading causes of death from cancer. Through early diagnosis and treatment, the survival rate of patients with breast cancer could be significantly improved. Automated breast ultrasound (ABUS) is the most common breast inspection technique and could provide three-dimensional (3-D) spatial information on breast tissue. However, ABUS images consisting of a large number of 2-D slices make radiologists spend much time reviewing and delaying the diagnosis of malignant tumors. Recently, deep learning-based computer-aided diagnosis (CADx) systems have been developed to extract features automatically and speed up the diagnostic process. Thus, this study proposed a CADx system based on the Transformer and convolutional neural network (CNN) for tumor diagnosis on ABUS images. The proposed tumor classification model employed the self-attention mechanism to capture the global relationship and added the inverted residual convolution block to extract local information from images.

The CADx system in this study contained data preprocessing, tumor segmentation, and tumor classification. In the data preprocessing, the tumor region was extracted as a volume of interest (VOI) and resized to the fixed image size. Afterward, the histogram equalization was conducted to increase the image contrast. Next, the resized VOI was utilized to generate a tumor mask through the tumor segmentation model. Finally, the

resized VOI, the enhanced VOI, and the corresponding tumor mask were used as the inputs of the proposed 3-D SWIRL Transformer tumor classification model to determine tumor malignancy. The proposed model was constructed with shifted window-based self-attention mechanism and inverted residual switch to aggregate global and local features. According to the results, the proposed CADx system could reach 89.9% accuracy, 89.8% sensitivity, 89.9% specificity, and 0.9401 AUC. These results demonstrated that the proposed CADx system could effectively help radiologists diagnose breast cancer more accurately.

Keywords: Breast cancer, Automated breast ultrasound (ABUS), Computer-aided diagnosis (CADx), Convolution neural network (CNN), Self-attention mechanism

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Chapter 1 Introduction

Breast cancer is the most common cancer and the leading cause of death compared to other cancer types for women worldwide [1]. The survival rate could be substantially improved through early detection, precise diagnosis, and proper treatment. Automated breast ultrasound (ABUS) is a feasible screening technique that reduces image acquisition time, decreases operator dependence, and enhances image reproducibility [2] for breast inspection. Besides, the ABUS is three-dimensional (3-D) data consisting of hundreds of 2-D slices and could provide more spatial information about the breast tissue. Nevertheless, the large number of images makes reviewing ABUS images time-consuming and leads to delays in diagnosing malignant tumors [3]. Meanwhile, radiologists' experience examining ABUS images can also influence the diagnosis. Therefore, computer-aided diagnosis (CADx) systems have recently been developed to facilitate reviewing AUBS images and help radiologists correctly diagnose breast tumors.

In 2018, Nemat *et al.* built a CADx system that employed mathematical transformation techniques to extract texture and morphological features and used the logistic regression classifier to diagnose tumors [4]. In 2019, Huang *et al.* proposed another CADx system that utilized fuzzy rules to make tumor diagnostic decisions based on the self-defined feature scores from the Breast Imaging Reporting and Data

System (BI-RADS) standard [5]. However, these conventional CADx systems only classified the tumors depending on the specific handcrafted features, making the diagnostic system performance limited by the selected features and vary from person to person. Lately, research on convolutional neural networks (CNNs) for medical image detection and classification tasks indicated that CADx systems with deep learning models could achieve convincing performance[6]. Unlike the conventional CADx systems, CNN-based CADx systems could automatically extract relevant features to determine whether the tumor was benign or malignant and have gradually become an option for radiologists to make more accurate diagnoses.

In the clinic, texture and morphological characteristics are vital information to help radiologists distinguish malignant tumors from benign lesions. However, the CNN-based CADx systems without lesion segmentation models only use texture features for tumor classification. Thus, current CADx systems usually have the tumor segmentation model that separates the tumor region from the background and other unrelated tissues to acquire masks for tumor diagnosis [4, 7]. In 2018, Zhou *et al.* proposed U-Net++ to perform segmentation on medical images by introducing nested dense convolutions to fuse features from different levels of the encoder-decoder structure [8]. Compared with U-Net, U-Net++ performed better on segmentation tasks of breast ultrasound images and breast dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI), with

9.1% and 4.82% improvement of intersection over union (IoU), respectively [9, 10].

These results suggested that U-Net++ could serve as a proper architecture for tumor segmentation and provide tumor masks for higher tumor classification performance.

Regarding the classification model, the Transformer-based models [11, 12] with the self-attention mechanism to obtain intra-data interactions achieved great success in image classification tasks. Unlike CNN, the Transformer-based model could capture long-range relationships regardless of the distance between image contents. In 2021, Swin Transformer was proposed to efficiently perform self-attention in a sliding window and construct a hierarchical structure for capturing different resolution features from different model stages [13]. The Swin Transformer achieved an outstanding performance of 84.5% in top-1 accuracy on the ImageNet image classification task, making it one of the state-of-the-art architectures. Since the clinical texture and morphological characteristics are essential to diagnose ultrasound breast tumors, Swin Transformer generating multi-scale features by patch merging could potentially classify tumors more correctly. However, the Transformer-based model focused more on the global dependencies of input data, and the extracted features might not be sufficient to represent the local relationships. Hence, combining the CNN into the Transformer-based model could be an intrinsic operation for aggregating local features into the model [14].

Considering the model parameters and computational burden, the lightweight CNN structure taking performance and efficiency into account played an essential role in extracting local features. In 2017, MobileNet containing depth-wise convolution was proposed to build lightweight models for embedded devices [15]. Next year, an inverted residual (IR) with the expansion layer in MobileNetV2 was developed based on the depth-wise convolution for memory efficiency and higher performance on image classification tasks [16]. According to their experiments, the MobileNetV2 with depth-wise convolution and inverted residual bottleneck layers was 3% higher in top-1 accuracy on the ImageNet task and had 22.4 million fewer parameters than the MobileNet with full convolution. Besides, the MobileNetV2, with 23.33 million fewer parameters than ResNet, could reach 93.17% accuracy on the classification task of the breast image dataset BreakHis [17]. These results demonstrated that the inverted residual bottleneck structure based on depth-wise convolution could effectively and efficiently collect more features by introducing the non-linear transformation of the features into the model. This CNN-based structure could be feasible to integrate more local features with the long-range relationships provided by the Transformer-based model.

Therefore, this study proposed a 3-D CADx with the tumor segmentation and classification model for ABUS tumor diagnosis. Among the proposed CADx system,

the tumor segmentation was performed by applying 3-D U-Net++, and the tumor classification was achieved by adopting the Swin Transformer as the backbone and employing the IR switch for tumor diagnosis. The followings were the details of the proposed CADx system. First, the materials, including the data source and tumor type of the dataset in this study, would be described in Chapter 2. Second, the workflow of the CADx system consisting of data preprocessing, tumor segmentation, and tumor classification was elaborated in Chapter 3. Third, the experiment results and discussion were shown in Chapter 4. Finally, the conclusion and future work were presented in Chapter 5.

Chapter 2 Materials

The 3-D ABUS dataset used in this study was provided by Sun Yat-sen University Cancer Center (SYSUCC) between June 2015 and March 2018 and was the same dataset employed by Xiang *et al.* [18]. InveniaTM Automated Breast Ultrasound System (GE Healthcare, Sunnyvale, CA, USA) with a frequency range from 6 to 14 MHz was used for taking these ABUS images. This dataset contains breast ultrasonic image data from 396 patients, and the average age of these participants was 49.2 ± 10.3 years. Among 444 pathology-confirmed tumors in the ABUS dataset, there were 218 benign and 226 malignant tumors. These benign and malignant tumors could be separated into more detailed tumor types. Among the benign tumors, there were 158 adenopathies, 43 fibroadenomas (FA), 12 intraductal papillomas (IDP), 4 sclerosing adenoses, and 1 mucinous carcinoma. As for the malignant lesions, there were 182 invasive ductal carcinomas (IDC), 28 ductal carcinomas in situ (DCIS), 10 invasive lobular carcinomas (ILC), 2 lobular carcinomas in situ (LCIS), 2 encapsulated papillary carcinomas (EPC), and 2 metaplastic carcinomas.

Besides, these tumors could also be divided into 406 mass and 38 non-mass tumors, according to the Breast Imaging Reporting and Data System (BI-RADS) [19]. A mass tumor is considered a space-occupying lesion on two different planes, whereas non-mass tumors lack an apparent shape or margin. The illustration of a mass and a non-

mass tumor was in Fig. 2-1. Non-mass tumors are about 9% of all breast tumors, and the malignancy rates of these tumors range from 26.8% to 58% [20-23]. Some ultrasound features between benign and malignant non-mass tumors are ambiguous, causing a high risk of misclassification and inappropriate clinical treatment. Therefore, this study also analyzed the performance of the proposed diagnostic model in discriminating between benign and malignant non-mass tumors. Complete information about the breast tumor dataset was organized in Table 2-1.

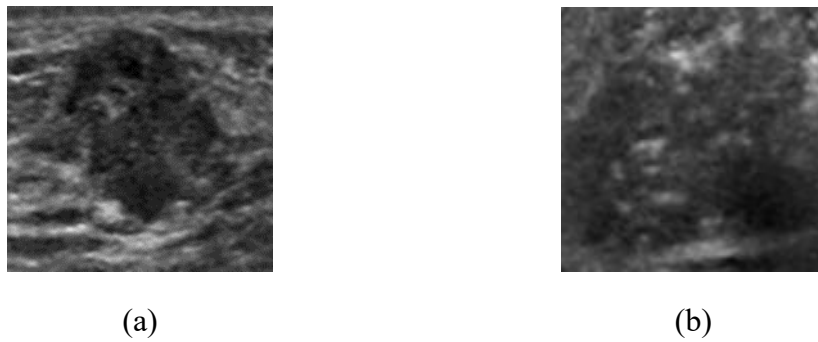


Fig. 2-1 Two different types of tumor, (a) a mass tumor and (b) a non-mass tumor.

Table 2-1 Tumor information of ABUS data.

Information about the patients			
Total number		396	
Mean age		49.2 ± 10.3	
Tumor type			
		Benign	Malignant
Mass tumor		203	203
Non-mass lesion		15	23
		218	226
Total		444	
Benign		Malignant	
Adenopathy	158	IDC	182
FA	43	DCIS	28
IDP	12	ILC	10
Sclerosing adenosis	4	LCIS	2
mucinous carcinoma	1	EPC	2
		metaplastic carcinoma	2

Chapter 3 Methods

In this study, a CADx system based on the improved Swin Transformer [13] was proposed for breast tumor classification. The CADx system was composed of data preprocessing, tumor segmentation, and tumor classification. In the data preprocessing, each tumor volume of interest (VOI) was manually marked by the radiologist and then extracted to reduce interference from background tissues. Next, the extracted VOIs were resized to the same shape to fit the proposed model's input size. Afterward, the histogram equalization was used on the resized VOIs to create a sufficient contrasted image. Meanwhile, the corresponding masks of the resized VOIs were generated by the tumor segmentation to provide the shape information of the tumors further. Finally, the resized VOIs, the contrasted VOIs, and the corresponding masks were concatenated as the input of the tumor classification model to predict whether a tumor was malignant or benign. The proposed CADx system flowchart was presented in Fig. 3-1 and introduced in the following sections.

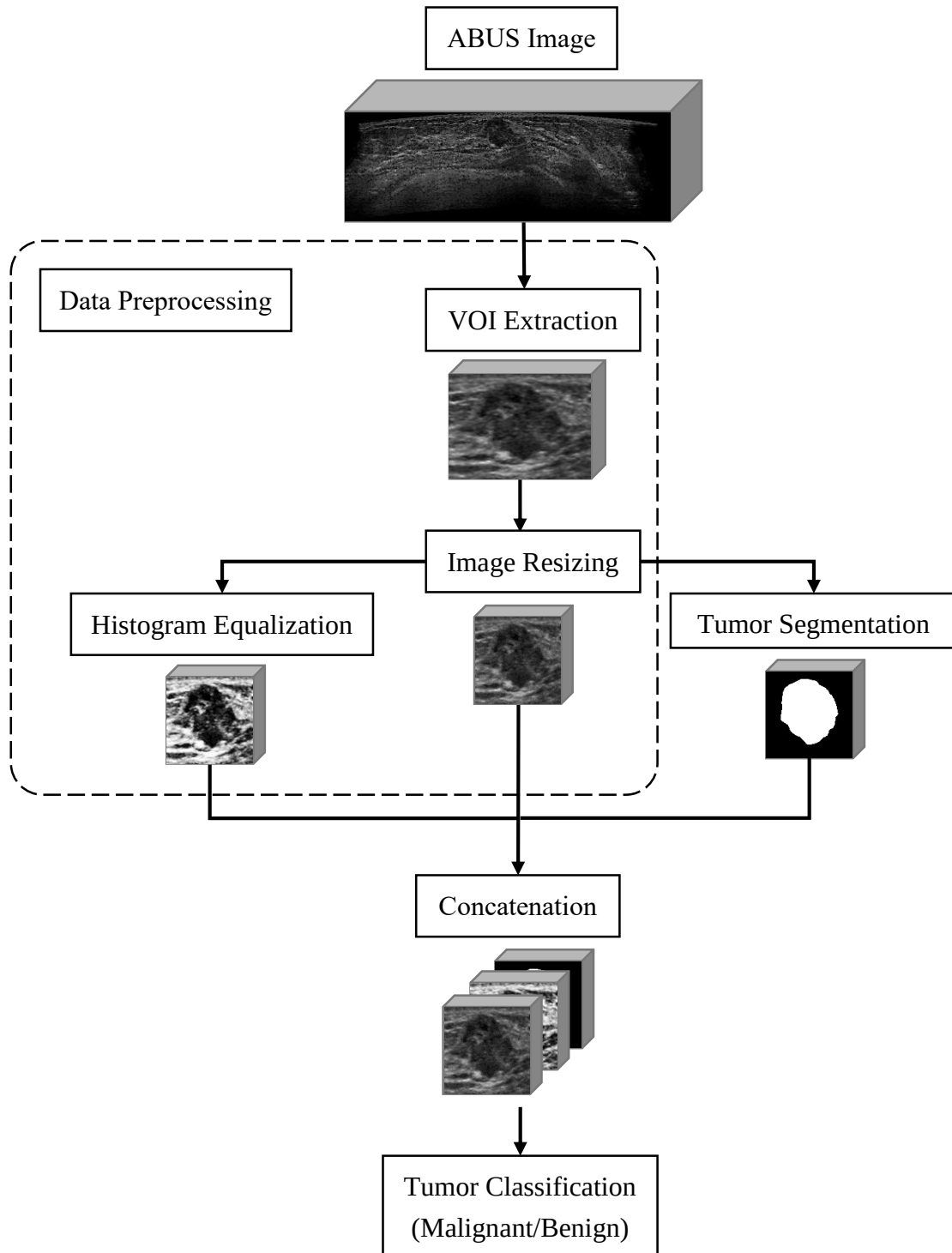


Fig. 3-1 The flowchart of the proposed CADx system.

3.1 Data Preprocessing

At the beginning of the proposed CADx system, data preprocessing comprised of VOI extraction, VOI resizing, and histogram equalization were performed sequentially

to enhance diagnostic model performance. First, the VOI extraction was performed to crop the tumor regions as VOI and reduce the background noises. Then considering the computational burden caused by 3-D data and adapting to the same input size of the proposed model, the extracted VOIs were resized to $16 \times 128 \times 128$ voxels by executing the VOI resizing. Afterward, since the dark and bright areas in the images might mix, the original VOI image would lack image contrast. Therefore, the contrast-enhanced images were produced through histogram equalization [24] to get more texture information about the tumors. The extracted VOI after the histogram equalization was shown in Fig. 3-2.

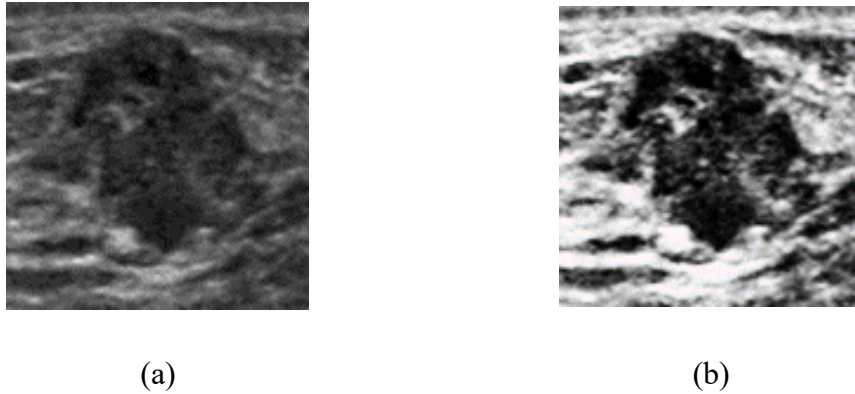


Fig. 3-2 (a) The extracted VOI and (b) the result after the histogram equalization.

3.2 Tumor Segmentation

After data preprocessing, the tumor masks would be obtained through tumor segmentation since the shape was a critical feature for tumor diagnosis. Hence, the segmented results would be part of the tumor classification model input to provide tumor shape information. This study applied the 3-D U-Net++ model [8] to acquire the

approximate tumor masks. Afterward, the segmented tumor images were post-processed to get more complete masks. Details of the tumor segmentation model and the post-processing were described in the following sections.

3.2.1 3-D U-Net++

In 2015, U-Net was first proposed by Ronneberger *et al.* [25] to achieve medical image segmentation. The U-Net was a U-shaped encoder-decoder network containing the contracting path for feature capture and the symmetric expansive path for context propagation to higher resolution layers. Besides, features from each convolution layer in the contracting path were cropped to prevent border pixel loss and concatenated with the corresponding up-sampled features in expanding path. However, two issues existed within the U-Net structure. The first issue was the optimal model depth would depend on variable datasets. The second issue was the direct skip connections of the feature maps from the contracting path and expansive path would cause semantic dissimilarity. Therefore, a nested U-Net, U-Net++ [8], was proposed to address the mentioned issues.

The structure of U-Net++ was similar to U-Net, consisting of the contracting path, the expanding path, and the specially designed skip connections. The illustration of the U-Net++ model was shown in Fig. 3-3. Different from U-Net directly concatenating the features in the contraction path with the same level features in the expansive path, the nested dense convolution structure in U-Net++ combined features from several

levels. Information of the objects to be segmented was captured by convolution layers and dense skip connections, enriching the details of the feature maps in the encoder network before fusing with up-sampled features in the decoder. Considering the mentioned advantages, U-Net++ was selected as our tumor segmentation model. Since the dataset in this study was 3-D ABUS images, the original U-Net++ was expanded from 2-D to 3-D convolutions, forming a 3-D U-Net++ model.

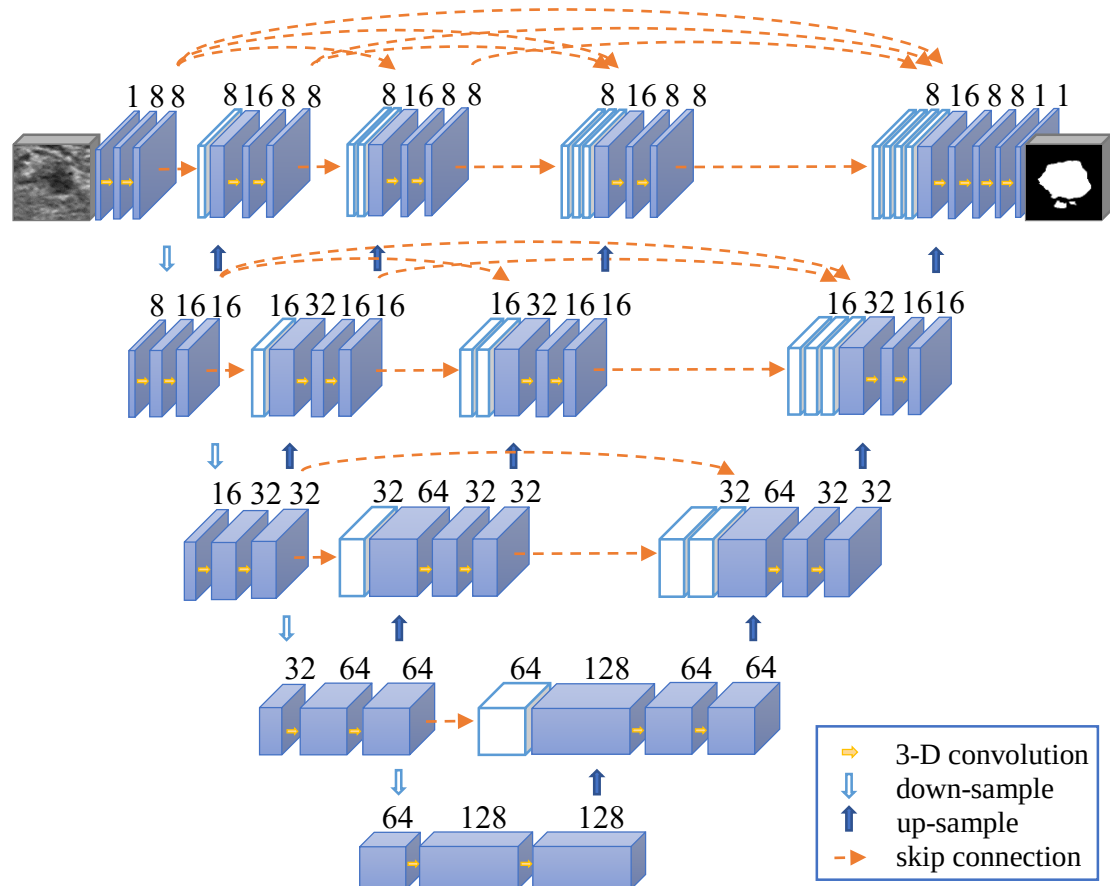


Fig. 3-3 The architecture of 3-D U-Net++.

3.2.2 Post-processing

After the tumor segmentation, the segmented results still needed some refinement to obtain finer-grained masks in the post-processing due to two issues. One was that a

few tumor voxels were possibly misclassified as background, shown in Fig. 3-4.

Another was that some tissues near the tumor were mis-segmented as the tumor regions, such as the Fig. 3-5 presentation. Hence, the hole-filling filter was adopted to correct the erroneous background voxels in the tumor regions, and the largest connected method was applied to remove the noises from non-tumor tissues. Fig. 3-4 and Fig. 3-5 illustrated the whole processes of tumor segmentation.

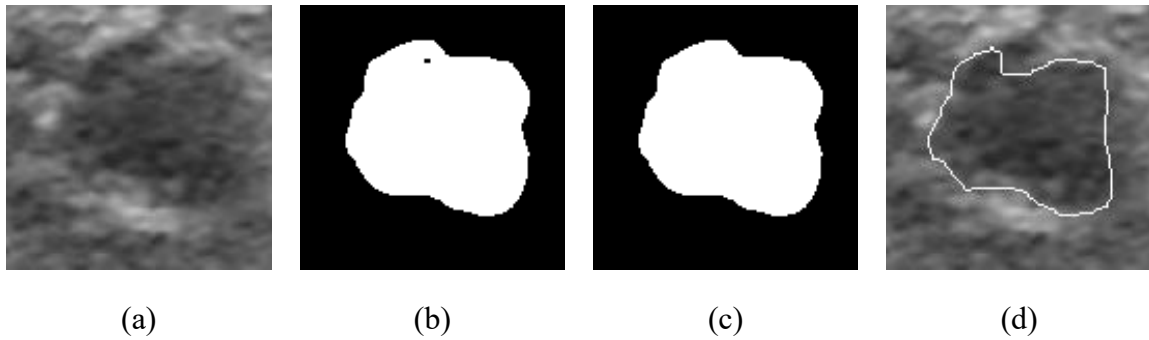


Fig. 3-4 Tumor segmentation of the mass tumor. (a) Input image. (b) The segmentation result with a hole. (c) The final segmentation result after performing the post-processing. (d) The ground truth contour.



Fig. 3-5 Tumor segmentation of the non-mass tumor. (a) Input image. (b) The segmentation result with mis-segmented tissues near the tumor regions. (c) The final segmentation result after performing the post-processing. (d) The ground truth contour.

3.3 Tumor Classification

After the tumor segmentation, the input data containing the resized VOIs, the enhanced VOI, and the VOI masks were delivered to our tumor classification model for tumor classification. Considering the used dataset, the limited computational resources, and classification performance, a hybrid architecture, 3-D SWIRL Transformer model modified from Swin Transformer [13] by the shifted window-based self-attention mechanisms and inverted residual (IR) structures [16], was proposed. Besides, a variety of data augmentation techniques were also employed to prevent the overfitting problem caused by insufficient training data amount [26, 27].

3.3.1 3-D SWIRL Transformer

Because our 3-D SWIRL Transformer was altered from Swin Transformer, the classification process was the same as Swin Transformer. The architecture of the 3-D

SWIRL Transformer model was shown in Fig. 3-6. The classification model included four stages for feature extraction, and a global average pooling (GAP) and a fully connected (FC) layer for the tumor classification. Moreover, the proposed SWIRL Transformer also adopted patch partition and patch merging to operate self-attention efficiently to image patches and extract multi-resolution features. Hence, two significant modifications could enhance model efficiency and model performance in our classification model. One was the embedding dimension of each image patch and the number of blocks would be reduced because our model was a 3-D model. Another was that the 3-D SWIRL Transformer block, developed by embedding the inverted residual switch (IR switch) module into the Swin Transformer block, replaced the Swin Transformer block to achieve better performance.

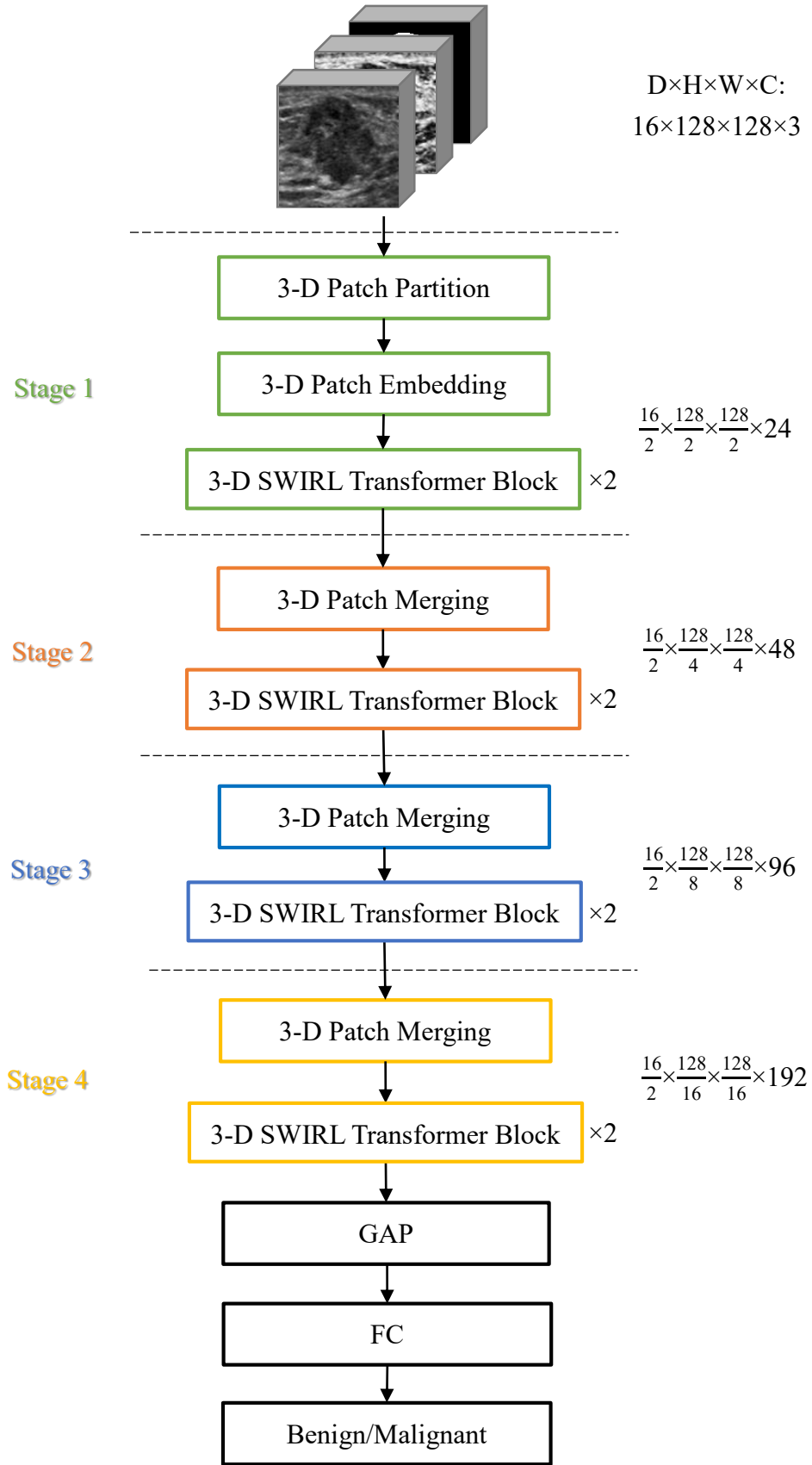


Fig. 3-6 The flowchart of the 3-D SWIRL Transformer.

3.3.2 3-D SWIRL Transformer block

In the proposed model, the features were generated by the SWIRL Transformer block at each stage. Like Swin Transformer blocks, there were two successive SWIRL Transformer blocks at every stage. The structure of the SWIRL Transformer block and Swin Transformer block was presented in Fig. 3-7. It was evident that the difference between the SWIRL and Swin Transformer block was the IR switch, which replaced the multilayer perceptron (MLP) of the Swin Transformer block. In two successive SWIRL Transformer blocks, most modules retained the designs in Swin Transformer blocks to maintain the advantages of the original blocks. These block designs contained the layer normalization (LN), the residual connections, the window-based multi-head self-attention (W-MSA) module, and shifted window-based multi-head self-attention (SW-MSA) module.

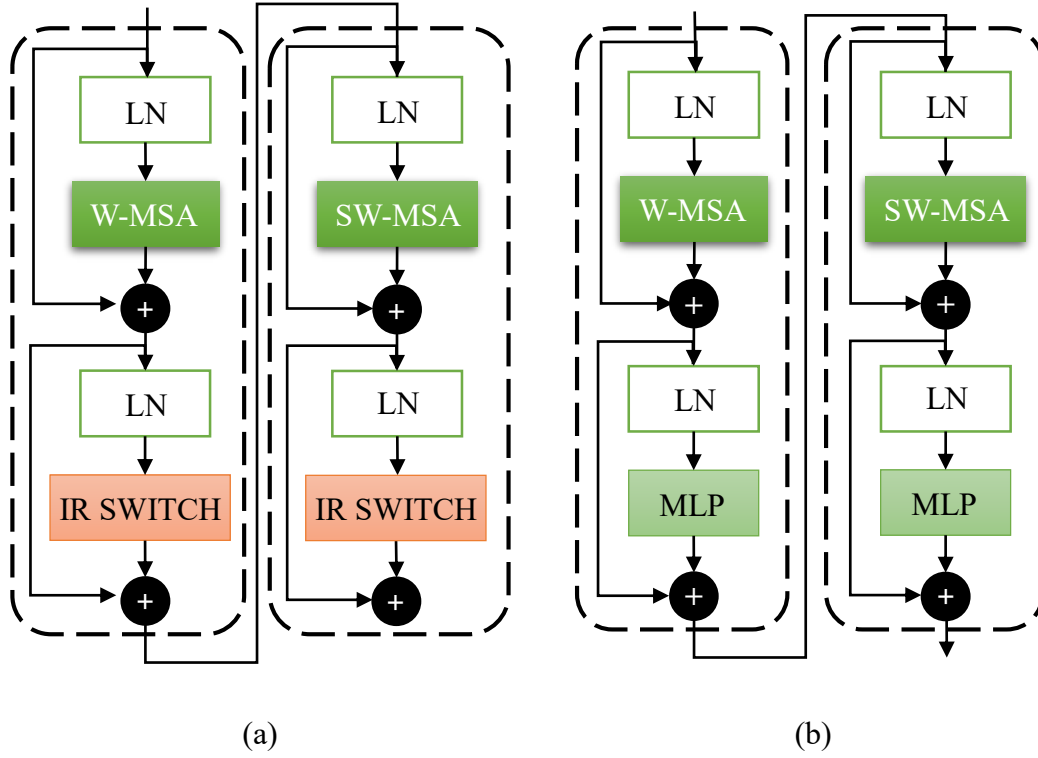


Fig. 3-7 (a) Two successive SWIRL-T blocks (b) Two successive Swin-T blocks.

W-MSA and SW-MSA modules were unique self-attention mechanisms that could reduce the computation cost. The workflow of the W-MSA and SW-MSA was shown in Fig. 3-8. In W-MSA, self-attention would only operate on the group of patches in a local sliding window way. Then, the SW-MSA was applied after the W-MSA by shifting the image patches to increase cross-window connections. Hence, the self-attention mechanism could operate on different combinations of image patches by the same local sliding window in W-MSA. Besides, window masks would also be generated to avoid self-attention to originally non-adjacent image patches. Although the data used in this study was the 3-D image, the shifting window method still maintained the way of the primary Swin Transformer. Next, the CNN-based structure, IR switch, was employed

to produce more texture information in the local receptive field. However, the features in the CNN and self-attention mechanism were distinct, and the feature format conversion would be executed for structure connection.

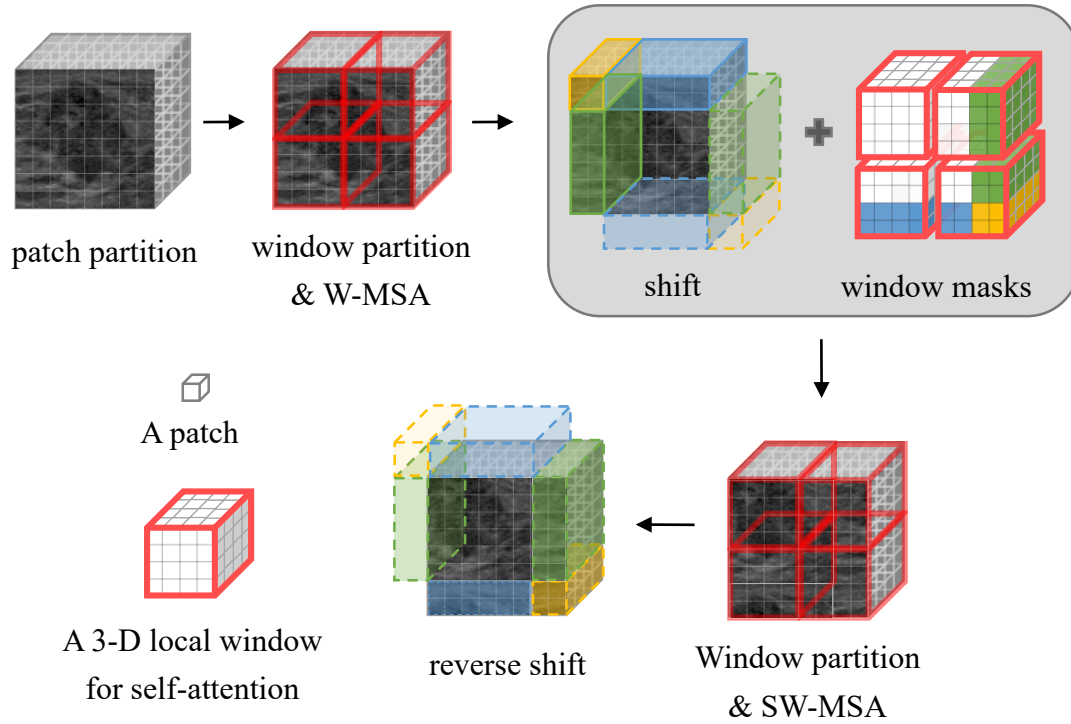


Fig. 3-8 Workflow of W-MSA and SW-MSA for 3-D ABUS images.

3.3.3 Inverted Residual Switch

To convert feature formats for structure connection, the switch mechanism would be employed for changing feature formats. In this study, the 3-D IR switch, presented in Fig. 3-9, was proposed to handle the mentioned problem. The 3-D IR switch contained the 3-D inverted residual block (3-D IR block) and two feature conversion layers, the Patch to Image layer and the Image to Patch layer. In IR switch, the Patch to

Image layer would first convert the feature format from the image patch format to a 3-D image format. Then, considering model computational efficiency, the 3-D IR block based on depth-wise convolution would be employed to obtain features. The 3-D IR block was a bottleneck structure while extracting features in a high dimension representation and transmitting features in a low dimension. Meanwhile, the CNN structure could acquire important local features for classification, which differed from the self-attention mechanism that focused more on global relationships between the image contents. After the feature extraction in IR-block, the Image to Patch layer could transfer the 3-D image format back to the image patch format.

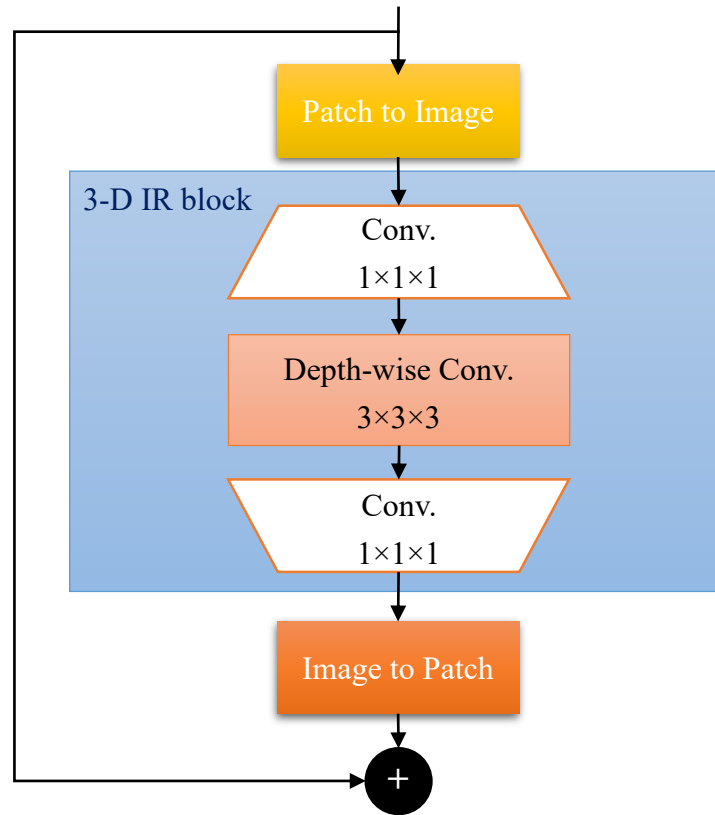


Fig. 3-9 The structure of the 3-D IR switch module.

3.3.4 Data Augmentation

Before training the proposed SWIRL Transformer model, due to the small-size dataset, the model would lack the generalization ability by training with insufficient data and result in the overfitting problem. Besides, while the model containing self-attention could attempt to learn locality by incorporating position embedding, the reduced inductive bias compared with the CNN-based model still caused the need for more data [28]. Therefore, data augmentation is a viable solution to the problem by generating many kinds of images in the training data [27]. This study implemented several data augmentation techniques, including flipping, random crop, cutout, mixup, and cutmix. In flipping, the 3-D input data was flipped in the vertical, horizontal, and depth directions. As for the random crop, the image would be first padded to a larger size and then randomly cropped to the size of $16 \times 128 \times 128$ voxels. In addition, several cuboid regions in the 3-D ABUS image would be randomly masked out by cutout to improve the model's regularization [29].

To further increase the diversity of the input data, mixup could generate virtual data by linear interpolations of the two examples in the training dataset [30]. The followings were the fusion equations between two random input examples:

$$\tilde{x}_{mixup} = \lambda_{mixup} x_i + (1 - \lambda) x_j \quad (1)$$

$$\tilde{y}_{mixup} = \lambda_{mixup} y_i + (1 - \lambda) y_j$$

where (x_i, y_i) and (x_j, y_j) were two examples and their labels selected from a batch of training data, λ_{mixup} was a factor and was randomly drawn from a Beta distribution [31] between 0 and 1. The Beta distribution is a continuous probability distribution defined by two positive parameters, α and β . In this study, the λ_{mixup} was randomly selected from the beta distribution with $\alpha = \beta = 0.5$.

In addition to the mixup augmentation strategy, the cutmix augmentation strategy would be also performed to generate more training data for a better-generalized network [32]. Cutmix would remove part of a training example and replace it with the same location region of another example. The combination operations were defined as follows:

$$\tilde{x}_{cutmix} = \mathbf{M} \odot x_i + (\mathbf{1} - \mathbf{M}) \odot x_j \quad (2)$$

$$\tilde{y}_{cutmix} = \lambda_{cutmix} y_i + (1 - \lambda_{cutmix}) y_j$$

where (x_i, y_i) and (x_j, y_j) were two training examples, $\mathbf{M} \in \{0,1\}^{D \times W \times H}$ was a binary matrix that masked out the removed region in one sample and substituted it with another example. $\mathbf{1}$ in bold was a binary matrix of size $D \times W \times H$ whose elements were all ones, and $\odot$ was the element-wise multiplication. λ_{cutmix} was a combination ratio between the two examples and was randomly sampled from the Beta distribution with $\alpha = \beta = 1$. In this study, only one of mixup and cutmix was implemented in a training batch, and the illustration of different data augmentation methods was in Fig. 3-10.

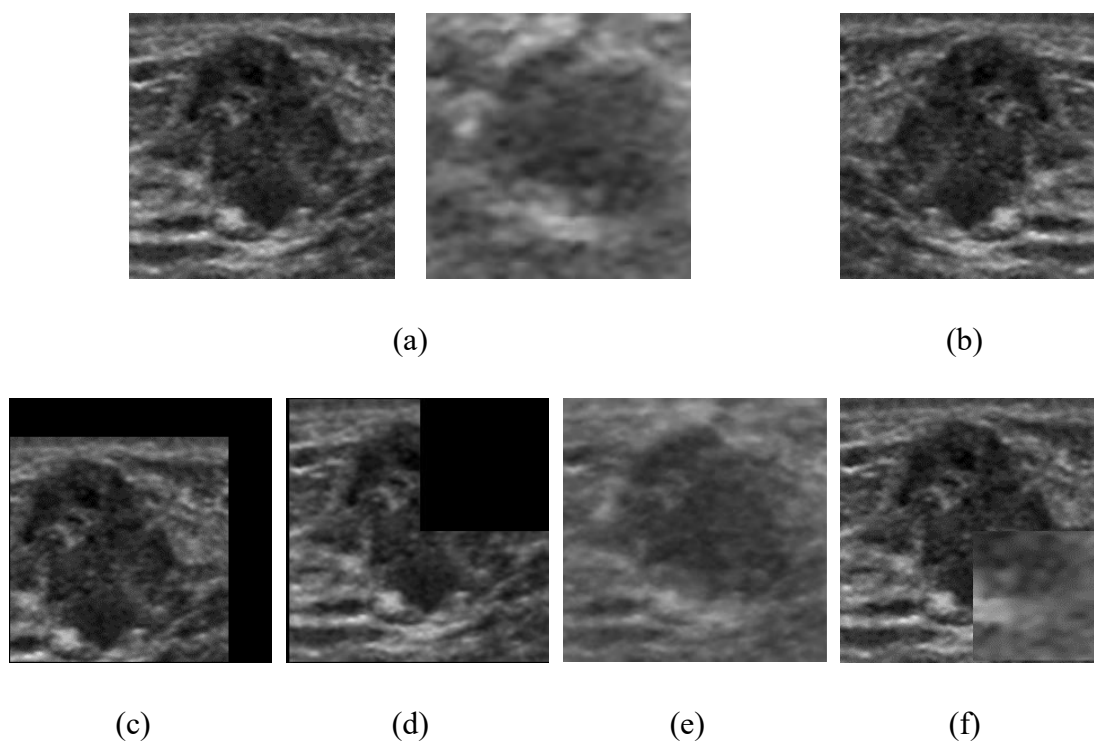


Fig. 3-10 The results of different data augmentation techniques. (a) Two tumor cases for data augmentation. (b) Flipping. (c) Random crop. (d) Cutout. (e) Mixup. (f) Cutmix.

Chapter 4 Experiment Results and Discussion

4.1 Experiment Environment

The proposed CADx system was built in an environment of the programming language Python 3.8.11. and the deep learning framework Pytorch 1.9.1. The computer used for experiments was equipped with Microsoft Windows 10 (Redmond, WA, USA). In terms of computer hardware, the experimental model and model analysis were implemented on an Intel Core i7-10700 2.9GHz CPU (NVIDIA, Santa Clara, CA, USA) with 16 GB RAM and an NVIDIA GeForce RTX 3090 GPU with 24 GB memory (NVIDIA, Santa Clara, CA, USA).

4.2 Evaluation

In this study, the proposed model was compared to other state-of-the-art deep learning models for CADx system performance evaluation. The experiments would conduct the 5-fold cross-validation [33] to verify the generalization ability of different models. The criteria for model comparison, as shown in (3), were accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and area under receiver operator characteristic curve (AUC). In addition, the p -value was calculated in McNemar's Test [34] and area test [35] to assess whether the differences between the proposed model and other models were statistically significant.

$$\begin{aligned}
\text{Accuracy} &= \frac{TP+TN}{TP+TN+FP+FN} \\
\text{Sensitivity} &= \frac{TP}{TP+FN} \\
\text{Specificity} &= \frac{TN}{TN+FP} \\
\text{Positive predictive value (PPV)} &= \frac{TP}{TP+FP} \\
\text{Negative predictive value (NPV)} &= \frac{TN}{TN+FN}
\end{aligned} \tag{3}$$

where TP, TN, FP, and FN represented true positive, true negative, false positive, and false negative.

4.3 Experiment Results

To evaluate the proposed CADx system, ablation study and model comparison were contained in the following experiments. In the ablation study, the validity of the downscaled model structure, IR switch module, and data augmentation techniques on the proposed model would be proven. In model comparison, the proposed model was compared with the previous research, Res-CapsNet [18] and six well-known models such as SE-Net [36], Inception [37], ResNet [38], ResNeXt [39], ResNeSt [40], and Swin Transformer [13]. The six famous models were extended from original 2-D models to 3-D models. In addition to the mentioned models, the diagnostic results of three experienced radiologists were also included. These professional radiologists, Radiologist 1, Radiologist 2, and Radiologist 3, have had 2, 3, and 5 years of ABUS diagnostic experience, respectively.

4.3.1 Ablation Study

To evaluate the validity of the IR switch, the SWIRL Transformer and the Swin Transformer were contained in the ablation study. Table 4-2 exhibited the model settings where the reduced structure was the modifications that decreased the embedding dimension of the image patch and the number of blocks compared to the primitive Swin Transformer. By reducing the size of the model structure, the number of model parameters could be decreased to prevent the overfitting problem and waste of computing resources. Table 4-2 and Fig. 4-1 presented the model performance and the corresponding receiver operating characteristic (ROC) curves. According to the results, the proposed model achieved the best scores among all the model settings, with 89.9% accuracy, 89.8% sensitivity, 89.9% specificity, 90.2% PPV, 89.5% NPV, and 0.9401 AUC. In conclusion, by adding the IR switch, the SWIRL Transformer generally performed better than the Swin Transformer. Furthermore, the reduced SWIRL Transformer scored higher than SWIRL Transformer in terms of accuracy, sensitivity, PPV, and NPV, with the same specificity but a slightly lower AUC. Moreover, the SWIRL Transformer with data augmentation could achieve better performance than SWIRL Transformer.

Table 4-1 The model settings of the ablation study

		Reduced Structure	Data Augmentation
Swin Transformer	<i>3-D Swin Transformer</i>		
	<i>Reduced 3-D Swin Transformer</i>	√	
	<i>3-D Swin Transformer with aug</i>		√
	<i>Reduced 3-D Swin Transformer with aug</i>	√	√
SWIRL Transformer	<i>3-D SWIRL Transformer</i>		
	<i>Reduced 3-D SWIRL Transformer</i>	√	
	<i>3-D SWIRL Transformer with aug</i>		√
	<i>Proposed 3-D SWIRL Transformer</i>	√	√

Table 4-2 The model performance of ablation study.

		Accuracy	Sensitivity	Specificity	PPV	NPV	AUC
Swin Transformer	<i>3-D Swin Transformer</i>	79.3% (352/444)	80.5% (182/226)	78.0% (170/218)	79.1% (182/230)	79.4% (170/214)	0.8282
	<i>Reduced 3-D Swin Transformer</i>	79.1% (351/444)	81.4% (184/226)	76.6% (167/218)	78.3% (184/235)	79.9% (167/209)	0.7930
	<i>3-D Swin Transformer with aug</i>	81.1% (360/444)	87.2% (197/226)	74.8% (163/218)	78.2% (197/252)	84.9% (163/192)	0.8668
	<i>Reduced 3-D Swin Transformer with aug</i>	83.3% (370/444)	86.3% (195/226)	80.3% (175/218)	81.9% (195/238)	85.0% (175/206)	0.8862
SWIRL Transformer	<i>3-D SWIRL Transformer</i>	84.0% (373/444)	85.8% (194/226)	82.1% (179/218)	83.3% (194/233)	84.8% (179/211)	0.8855
	<i>Reduced 3-D SWIRL Transformer</i>	85.1% (378/444)	88.1% (199/226)	82.1% (179/218)	83.6% (199/238)	86.9% (179/206)	0.8840
	<i>3-D SWIRL Transformer with aug</i>	88.7% (394/444)	88.5% (200/226)	89.0% (194/218)	89.3% (200/224)	88.2% (194/220)	0.9263
	<i>Proposed 3-D SWIRL Transformer</i>	89.9% (399/444)	89.8% (203/226)	89.9% (196/218)	90.2% (203/225)	89.5% (196/219)	0.9401

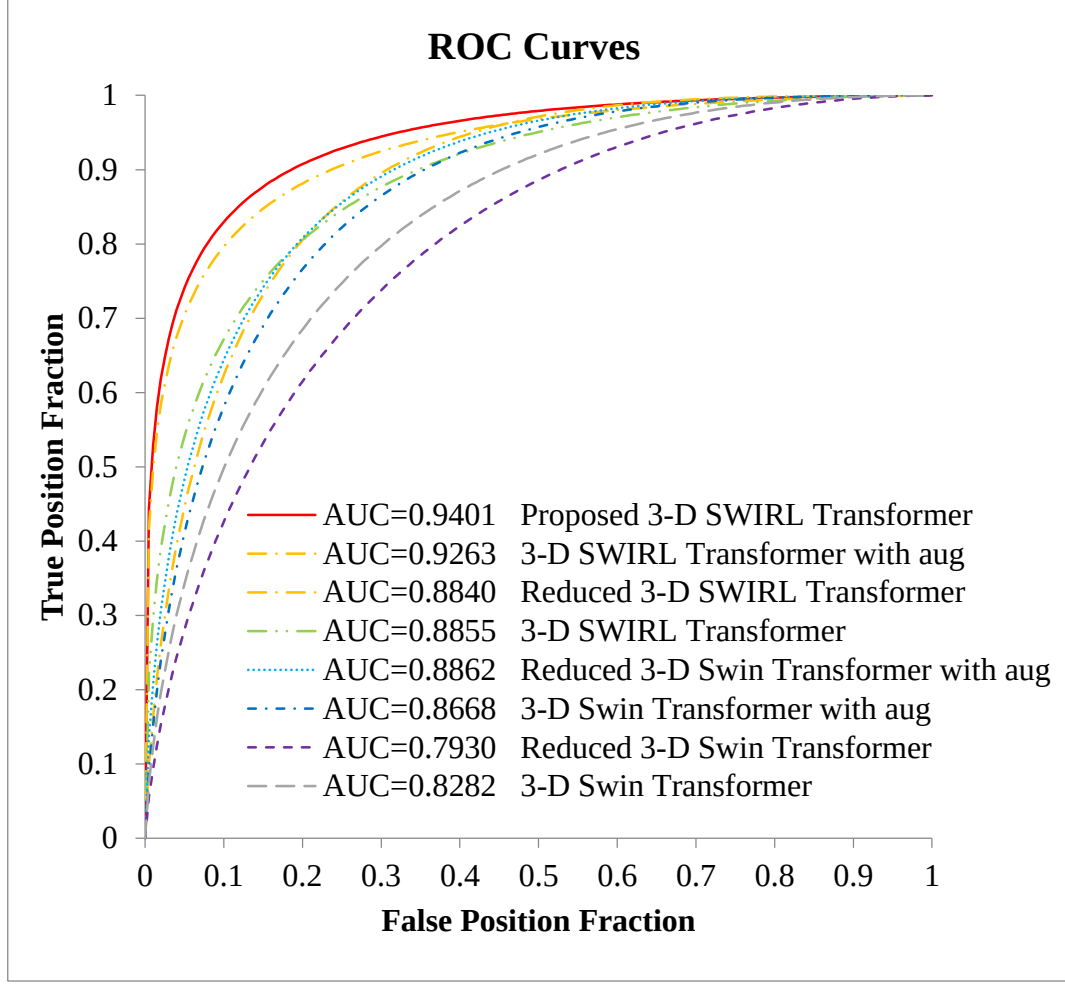


Fig. 4-1 The ROC curves in the ablation study.

4.3.2 Model Comparison

After the ablation study, the model comparison was implemented to evaluate the performance of the proposed SWIRL Transformer and other models. The diagnostic results of 3-D models and radiologists were listed in Table 4-3. To access the differences between 3-D and 2-D models, the 2-D version of the proposed model, 2-D CapsNet [41], and six mentioned deep learning models were also listed in the experiment results. These 2-D models would be trained by the key slice extracted from each 3-D ABUS data. Overall, the results demonstrated that 3-D models generally outperformed 2-D

models. As shown in Table 4-3, the proposed model reached 89.9% accuracy, 89.8% sensitivity, 89.9% specificity, 90.2% PPV, and 89.5% NPV, achieving the highest scores among all the 3-D and 2-D models. Compared to experienced radiologists, the performance of the proposed model was superior to Radiologist 1 and Radiologist 2 but inferior to Radiologist 3. Besides, the ROC curves of all the 3-D and 2-D models were plotted in Fig. 4-2 and Fig. 4-3. The proposed model achieved an AUC score of 0.9401, the highest among all the models.

Table 4-3 The results of all 3-D models, 2-D models, and radiologists in all tumors.

	Accuracy	Sensitivity	Specificity	PPV	NPV	AUC
3-D models						
<i>3-D SE-Net</i>	86.0% (382/444)	85.4% (193/226)	86.7% (189/218)	86.9% (193/222)	85.1% (189/222)	0.9152
<i>3-D Inception</i>	85.6% (380/444)	82.7% (187/226)	88.5% (193/218)	88.2% (187/212)	83.2% (193/232)	0.9175
<i>3-D ResNet</i>	85.6% (380/444)	87.2% (197/226)	83.9% (183/218)	84.9% (197/232)	86.3% (183/212)	0.9117
<i>3-D ResNeXt</i>	84.7% (376/444)	86.3% (195/226)	83.0% (181/218)	84.1% (195/232)	85.4% (181/212)	0.9111
<i>3-D ResNeSt</i>	87.2% (387/444)	87.2% (197/226)	87.2% (190/218)	87.6% (197/225)	86.8% (190/219)	0.9364
<i>3-D Res-CapsNet</i>	84.9% (377/444)	87.2% (197/226)	82.6% (180/218)	83.8% (197/235)	86.1% (180/209)	0.9122
<i>3-D Swin Transformer</i>	83.3% (370/444)	86.3% (195/226)	80.3% (175/218)	81.9% (195/238)	85.0% (175/206)	0.8862
<i>Proposed 3-D SWIRL Transformer</i>	89.9% (399/444)	89.8% (203/226)	89.9% (196/218)	90.2% (203/225)	89.5% (196/219)	0.9401

2-D models						
<i>2-D SE-Net</i>	79.7% (354/444)	81.8% (185/226)	77.5% (169/218)	79.1% (185/234)	80.5% (169/210)	0.8449
<i>2-D Inception</i>	72.5% (322/444)	72.6% (164/226)	72.5% (158/218)	73.2% (164/224)	71.8% (158/220)	0.7924
<i>2-D ResNet</i>	75.0% (333/444)	77.9% (176/226)	72.0% (157/218)	74.3% (176/237)	75.9% (172/207)	0.7959
<i>2-D ResNeXt</i>	77.0% (342/444)	74.3% (168/226)	79.8% (174/218)	79.3% (168/212)	75.0% (174/232)	0.7926
<i>2-D ResNeSt</i>	78.4% (348/444)	77.4% (175/226)	79.4% (173/218)	79.6% (175/220)	77.2% (157/224)	0.8450
<i>2-D CapsNet</i>	75.9% (337/444)	89.8% (203/226)	61.5% (134/218)	70.7% (203/287)	85.4% (134/157)	0.8000
<i>2-D Swin Transformer</i>	77.9% (346/444)	83.2% (188/226)	72.5% (158/218)	75.8% (188/248)	80.6% (158/196)	0.8101
<i>2-D SWIRL Transformer</i>	80.0% (355/444)	81.0% (183/226)	78.9% (172/218)	79.9% (183/229)	80.0% (172/215)	0.8534
Radiologists						
<i>Radiologist 1</i>	82.2% (365/444)	83.2% (188/226)	81.2% (177/218)	82.1% (188/229)	82.3% (177/215)	-
<i>Radiologist 2</i>	88.1% (391/444)	89.8% (203/226)	86.2% (188/218)	87.1% (203/233)	89.1% (188/211)	-
<i>Radiologist 3</i>	93.7% (416/444)	97.8% (221/226)	89.5% (195/218)	90.6% (221/244)	97.5% (195/200)	-

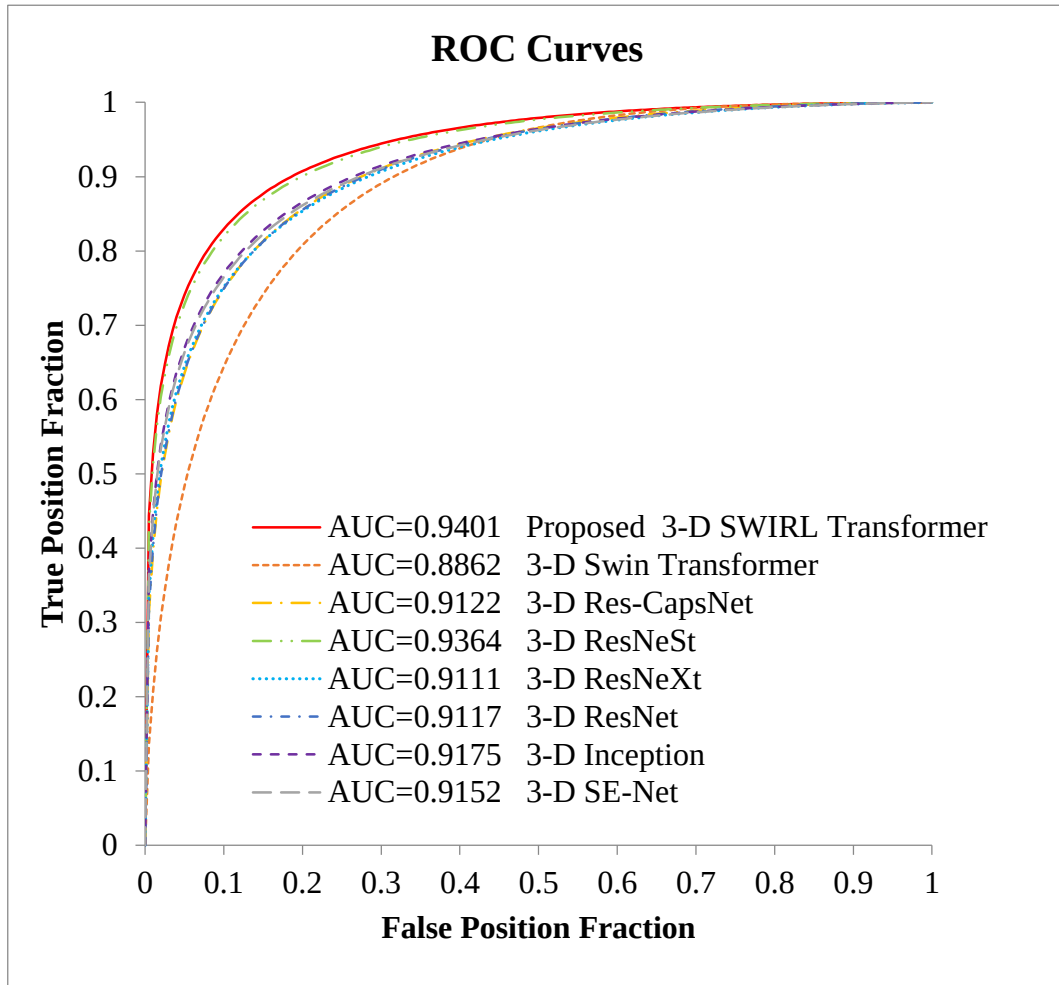


Fig. 4-2 The ROC curves of different 3-D models in all tumors.

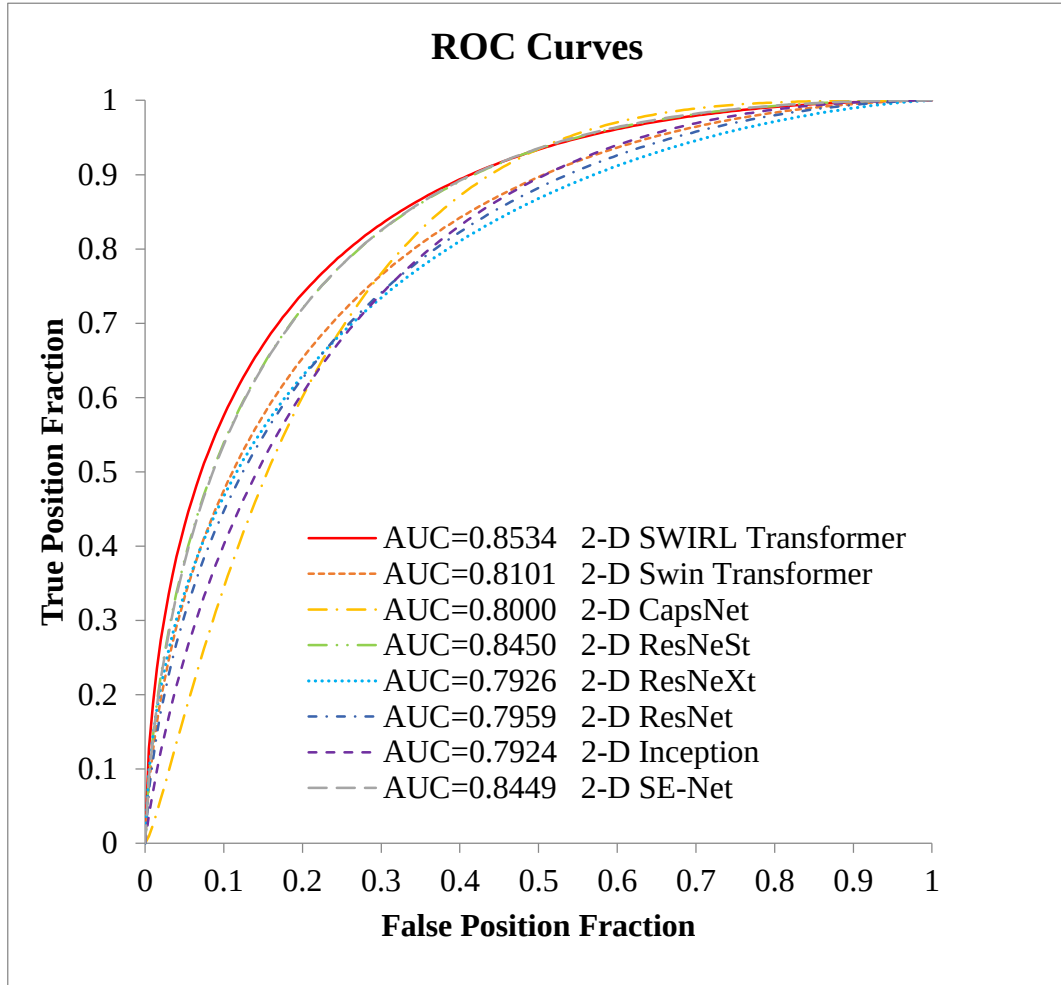


Fig. 4-3 The ROC curves of different 2-D models in all tumors.

Furthermore, the p -values derived from McNemar's test and area test were listed in Table 4-4, estimating whether significant differences existed between the proposed model and other 3-D models. The significance threshold was set as 0.05, and the p -value smaller than the threshold represented a significant difference in statistics. As shown in Table 4-4, the proposed model was statistically significantly different from all 3-D compared models except 3-D ResNeSt.

Table 4-4 The p -values for the proposed model and other models in all tumors.

	McNemar's test	Area test
<i>3-D SE-Net</i>	0.0196*	0.0083*
<i>3-D Inception</i>	0.0171*	0.0352*
<i>3-D ResNet</i>	0.0073*	0.0025*
<i>3-D ResNeXt</i>	0.0013*	0.0026*
<i>3-D ResNeSt</i>	0.0973	0.6897
<i>3-D Res-CapsNet</i>	0.0077*	0.0196*
<i>3-D Swin Transformer</i>	0.0004*	0.0001*

* The difference was statistically significant (p -value less than 0.05).

To further assess the performance of the proposed model, the predictive results of mass and non-mass tumors were analyzed separately. For mass tumors, the experiment results of different 3-D models were presented in Table 4-5. Among all the 3-D models, the proposed model achieved the highest performance in all the indices. The AUC of the proposed model also obtained the best score of 0.9457, as illustrated in the ROC curves of Fig. 4-4. According to Table 4-6, the proposed model differed significantly from other 3-D models, except for Inception on area test and ResNeSt on McNemar's and area tests.

Table 4-5 The results of different 3-D models in mass tumors.

	Accuracy	Sensitivity	Specificity	PPV	NPV	AUC
<i>3-D SE-Net</i>	86.9% (353/406)	86.7% (176/203)	87.2% (177/203)	87.1% (176/202)	86.8% (177/204)	0.9191
<i>3-D Inception</i>	86.2% (350/406)	83.7% (170/203)	88.7% (180/203)	88.1% (170/193)	84.5% (180/213)	0.9294
<i>3-D ResNet</i>	86.5% (351/406)	88.7% (180/203)	84.2% (171/203)	84.9% (180/212)	88.1% (171/194)	0.9190
<i>3-D ResNeXt</i>	85.2% (346/406)	86.7% (176/203)	83.7% (170/203)	84.2% (176/209)	86.3% (170/197)	0.9151
<i>3-D ResNeSt</i>	87.7% (356/406)	87.7% (178/203)	87.7% (178/203)	87.7% (178/203)	87.7% (178/203)	0.9409
<i>3-D Res-CapsNet</i>	85.2% (346/406)	88.2% (179/203)	82.3% (167/203)	83.3% (179/215)	87.3% (167/191)	0.9147
<i>3-D Swin Transformer</i>	83.7% (340/406)	87.2% (177/203)	80.3% (163/203)	81.6% (177/217)	86.2% (163/189)	0.8920
<i>Proposed 3-D SWIRL Transformer</i>	90.6% (368/406)	90.1% (183/203)	91.1% (185/203)	91.0% (183/201)	90.2% (185/205)	0.9457

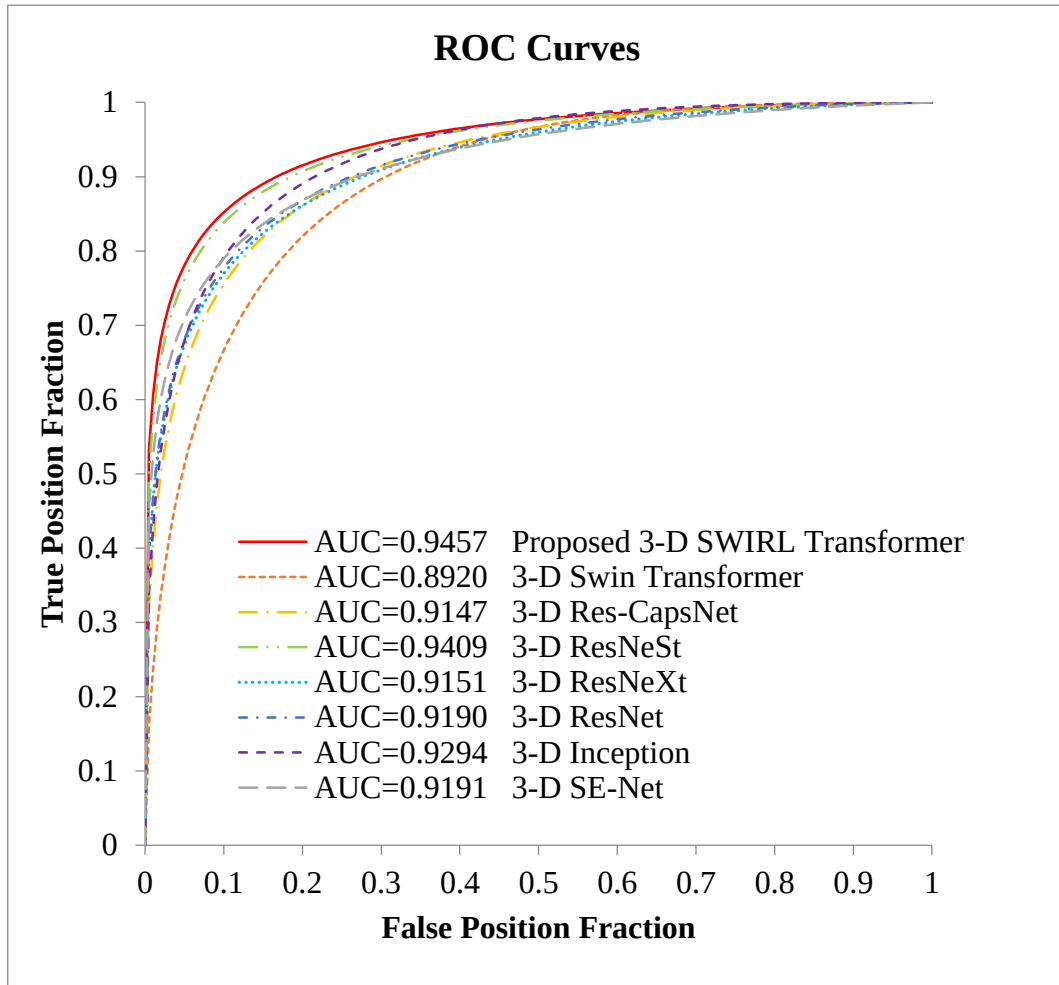


Fig. 4-4 The ROC curves of different 3-D models in mass tumors.

Table 4-6 The p -values for the proposed model and other models in mass tumors.

	McNemar's test	Area test
<i>3-D SE-Net</i>	0.0449*	0.0046*
<i>3-D Inception</i>	0.0251*	0.1262
<i>3-D ResNet</i>	0.0177*	0.0040*
<i>3-D ResNeXt</i>	0.0023*	0.0018*
<i>3-D ResNeSt</i>	0.1093	0.6403
<i>3-D Res-CapsNet</i>	0.0070*	0.0105*
<i>3-D Swin Transformer</i>	0.0006*	0.0001*

* The difference was statistically significant (p -value less than 0.05).

Compared with the mass tumors, the number of non-mass tumors in the dataset was small. Hence, the data information might not be enough for the model to extract specific features in non-mass tumors, causing the model performance in non-mass tumors to be worse than in mass tumors. The experiment results in non-mass tumors were exhibited in Table 4-7. Compared to other 3-D models, our SWIRL Transformer model only accomplished the highest scores of 81.6% and 87.0% in accuracy and sensitivity, respectively. For ROC curves demonstrated in Fig. 4-5, the AUC score of the proposed model was 0.8231, generally lower than other models. In addition, the p -values in Table 4-8 showed no statistically significant difference between the proposed model and other models on McNemar's test. However, on the area test, the proposed model was significantly different from other 3-D models except for ResNeSt.

Table 4-7 The results of different 3-D models in non-mass tumors.

	Accuracy	Sensitivity	Specificity	PPV	NPV	AUC
<i>3-D SE-Net</i>	76.3% (29/38)	73.9% (17/23)	80.0% (12/15)	85.0% (17/20)	66.7% (12/18)	0.8464
<i>3-D Inception</i>	78.9% (30/38)	73.9% (17/23)	86.7% (13/15)	89.5% (17/19)	68.4% (13/19)	0.8022
<i>3-D ResNet</i>	76.3% (29/38)	73.9% (17/23)	80.0% (12/15)	85.0% (17/20)	66.7% (12/18)	0.8681
<i>3-D ResNeXt</i>	78.9% (30/38)	82.6% (19/23)	73.3% (11/15)	82.6% (19/23)	73.3% (11/15)	0.8484
<i>3-D ResNeSt</i>	81.6% (31/38)	82.6% (19/23)	80.0% (12/15)	86.4% (19/22)	75.0% (12/16)	0.8839
<i>3-D Res-CapsNet</i>	81.6% (31/38)	78.3% (18/23)	86.7% (13/15)	90.0% (18/20)	72.2% (13/18)	0.8654
<i>3-D Swin Transformer</i>	78.9% (30/38)	78.3% (18/23)	80.0% (12/15)	85.7% (18/21)	80.0% (12/15)	0.8098
<i>Proposed 3-D SWIRL Transformer</i>	81.6% (31/38)	87.0% (20/23)	73.3% (11/15)	83.3% (20/24)	78.6% (11/14)	0.8231

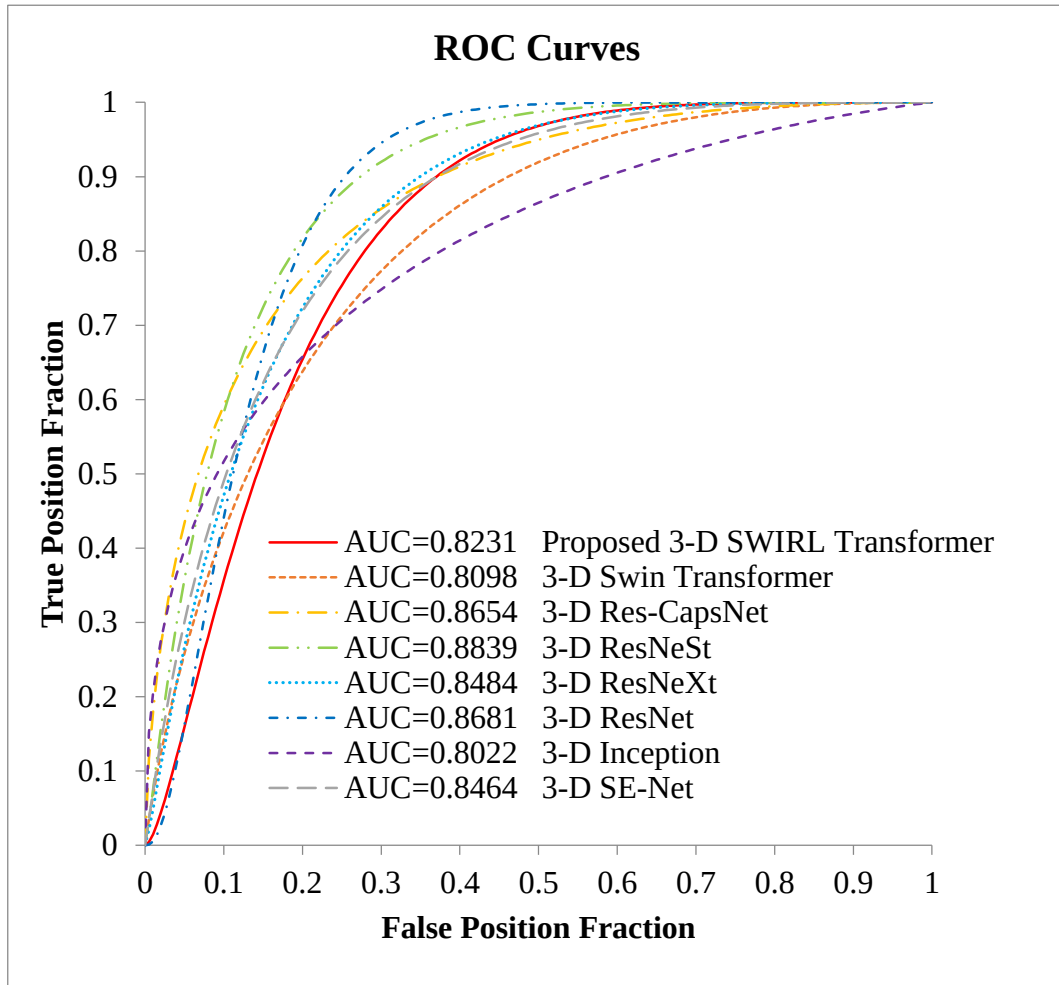


Fig. 4-5 The ROC curves of different 3-D models in non-mass tumors.

Table 4-8 The p -values for the proposed model and other models in non-mass tumors.

	McNemar's test	Area test
<i>3-D SE-Net</i>	0.3711	0.0083*
<i>3-D Inception</i>	0.6831	0.0352*
<i>3-D ResNet</i>	0.3711	0.0025*
<i>3-D ResNeXt</i>	0.6171	0.0026*
<i>3-D ResNeSt</i>	1.0000	0.6897
<i>3-D Res-CapsNet</i>	1.0000	0.0196*
<i>3-D Swin Transformer</i>	0.6831	0.0001*

* The difference was statistically significant (p-value less than 0.05).

4.4 Discussion

In this study, the CADx system constructed with the data preprocessing, the 3-D tumor segmentation model, and the 3-D tumor classification model was proposed for tumor classification in the ABUS image. In the data preprocessing, the VOIs only containing tumor and surrounding tissues would be cropped and resized to the fixed $16 \times 128 \times 128$ voxels. Next, the resized VOIs would be processed through the histogram equalization to get the contrasted images. Then, the 3-D U-Net++ was employed to generate the tumor masks in the tumor segmentation. Finally, the preprocessed VOIs and the corresponding masks were fed into the tumor classification model. In the tumor classification, the 3-D SWIRL Transformer, a hybrid architecture with the shifted window-based self-attention mechanism and IR switch, was employed to determine tumor malignancy. To evaluate the effectiveness of the proposed model, the ablation study and model comparison were performed. In the ablation study shown in Table 4-2, the implementation of the IR switch resulted in the overall performance of the SWIRL Transformer outperforming the Swin Transformer. Besides, the reduced structure or data augmentation could enhance almost all the metrics, allowing the proposed 3-D SWIRL Transformer to achieve the most precise tumor diagnosis. As for the model comparison in Table 4-3, 3-D models performed better than 2-D models, which indicated that the lesion depth, 3-D spatial relationship, and geometry information

provided by the 3-D data effectively improved model performance. These results demonstrated that the proposed 3-D SWIRL Transformer with 89.9% accuracy, 89.8% sensitivity, 89.9% specificity, 90.2% PPV, 89.5% NPV, and 0.9401 AUC outperformed other models in each evaluation metric. Besides, the performance of the proposed model was compared with three professional radiologists. The results of the proposed model were more excellent than the 2 and 3-year experienced radiologists but worse than the 5-year experienced radiologist.

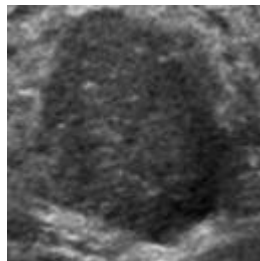
Due to the importance of the clinical texture and morphological characteristics, our previous research on the diagnosis of breast tumors also employed deep learning models to extract relevant features. In 2020, Xiang *et al.* used the same ABUS dataset to propose a 3-D CADx system, including a 3-D U-Net segmentation model and a 3-D Res-CapsNet classification model, for breast tumor classification [18]. In this research, the 3-D Res-CapsNet classification model was constructed with the capsule net (CapsNet) and the residual block (ResBlock) to capture more spatial information [41] and enrich different feature levels [38]. According to the results, the 3-D Res-CapsNet could reach an accuracy, sensitivity, specificity, PPV, NPV, and AUC of 84.9%, 87.2%, 82.6%, 83.8%, 86.1%, and 0.9122. In addition, several novel deep learning architectures were expanded to 3-D structures and included in the comparison. In Table 4-3, the 3-D Swin Transformer that was adopted as the backbone of the proposed model

failed to compete with other state-of-art CNN architectures, suggesting that the inductive bias advantageous to diagnosis was not adequately captured in the local receptive field. Compared with different classification models, the proposed 3-D SWIRL Transformer utilized W-MSA to perform the self-attention efficiently and capture the long-range dependency of image content. Moreover, by shifting the image patches, SW-MSA could increase the information exchange and strengthen the spatial relationship between different windows. Furthermore, patch merging that increased the scale of image patches at different stages could construct a hierarchical structure, gradually enlarging neurons' receptive field in the deeper network and obtaining multi-scale features. Besides, the IR switch in the proposed model could extract local features between spatially nearby and related voxels, thus, getting more inductive bias in the local receptive field [42]. Meanwhile, the feature expansion in the IR switch enabled the non-linear transformation of features and kept more key information for tumor classification. Therefore, the SWIRL Transformer introducing IR structure could integrate both short-range and long-range relationships of image contents. Combining the mentioned structure designs, the performance of the proposed model could surpass the 3-D Res-CapsNet and reach the best results on the ABUS classification task in all metrics.

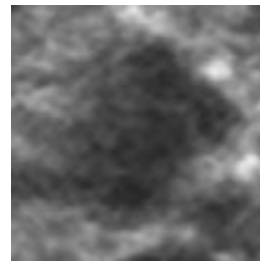
In clinical ultrasound imaging, the malignant lesions usually have the

characteristics of spiculated or angular margin, micro-calcification, and posterior shadowing, whereas the features of benign lesions are circumscribed margin and uniform echotexture [43, 44]. However, as shown in Fig. 4-6 (a), the malignant mass tumor with circumscribed margin and micro-calcification was misclassified as a benign tumor by three radiologists. Another case shown in Fig. 4-6 (b) was a benign non-mass tumor having an irregular margin and was considered malignant by all the radiologists. The characteristics of these two cases were too ambiguous to be judged by the common definition and confused the diagnosis of experienced radiologists. Therefore, these results provided that the proposed model could potentially assist radiologists in tumor classification in the future. In addition, non-mass lesions that do not meet the definition of mass tumor in BI-RADS are rare but still have the probability of malignancy and lack a diagnostic standard for non-mass tumors [45]. In 2021, Zhang *et al.* [46] using binary logistic regression on breast contrast-enhanced ultrasound (CEUS) could only obtain the AUC of 0.775 for non-mass lesion diagnosis, and the result using the combination of CEUS and BI-RADS could reach 0.885 in AUC. The result suggested that diagnosing non-mass tumors using breast ultrasound images alone was not easy. In addition, current research using a deep learning model on tumor diagnosis usually lacks evaluation related to non-mass tumors. Thus, it is worthwhile to subdivide all tumors into mass and non-mass tumors to investigate the effect of the proposed model on mass

and non-mass tumors, respectively. The experiment results of all the 3-D models in mass and non-mass tumors were shown in Table 4-5 and Table 4-7. The proposed model in mass tumors could reach 90.6% accuracy, 90.1% sensitivity, 91.1% specificity, 91.0% PPV, 90.2% NPV, and 0.9457 AUC, which was the highest performance among all the 3-D models. On the other hand, the results of the proposed model in non-mass tumors only attained the best scores of 81.6% and 87.0% in accuracy and sensitivity. The possible reason was that the number of non-mass tumors was too small to extract sufficient detailed features of non-mass tumors from all compared deep learning models, making it hard for the models to determine the tumors as malignant or benign [28].



(a)



(b)

Fig. 4-6 The breast tumors that all the radiologists misdiagnosed, (a) a malignant mass tumor, and (b) a benign non-mass tumor.

Even though the proposed model had great diagnostic results in most evaluations, there were still shortcomings in our model. The first was that the performance of the proposed model in non-mass lesions could not reach the best, causing less robustness

in predicting non-mass tumor malignancy. The way to deal with this problem was to enlarge the number of non-mass tumors in the dataset or adopt a novel loss function that could handle the data imbalance, such as real-world-weight cross-entropy loss (RWWCE) [47]. The second was the lack of channel attention in the proposed model, which could be solved by employing multi-spectral channel attention [48]. Different from squeeze-and-excitation block using global average pooling, Qin *et al.* [48] chose discrete cosine transformation (DCT) to generate a more diverse channel-wise feature selection scheme. The third was that the information of tumor size in anteroposterior, sagittal, and transverse dimensions was lost after VOI resizing. This factor would result in the misjudgment of the proposed model because the malignant tumors usually have the larger anteroposterior dimension than either the transverse or sagittal dimension [43]. By combining the existing features and tumor size information, the gradient boosting machine (GBM) [49] might be an efficient and effective method to apply several weak learners for finer classification.

Chapter 5 Conclusion

In this study, a CADx system comprised of data preprocessing, a 3-D U-Net++ tumor segmentation model, and a 3-D SWIRL Transformer tumor classification model was proposed for ABUS breast tumor diagnosis. First, the data preprocessing, including the VOI extraction, VOI resizing, and histogram equalization, was executed to generate VOI images and contrast-enhanced images. Then, the 3-D U-Net++ segmentation model and post-processing were employed to produce the corresponding tumor masks from the VOI images. Afterward, the proposed 3-D SWIRL Transformer model was utilized to classify the tumor as malignant or benign. Eventually, the proposed model attained the highest performance with 90.6% accuracy, 90.1% sensitivity, and 91.1% specificity. Besides, our CADx system reached 90.6% accuracy, 90.1% sensitivity, 91.1% specificity in mass tumors and 81.6% accuracy, 87.0% sensitivity, 73.3% specificity in non-mass tumors.

Although the proposed CADx system had excellent diagnostic capabilities, there was still room for improvement. First, more ABUS data should be collected, or a more proper loss function such as RWWCE should be applied to make the breast tumor diagnostic system more robust. Second, the feature diversity could be increased by channel attention methods such as multi-spectral channel attention. Third, GBM might integrate more tumor information into the tumor classification model for better

diagnostic results. In the future, the mentioned methods have the potential to overcome the limitations existing in the proposed CADx system.

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